

Section I: Data Structures [15 Marks]

Q1) Read the data from the file “employee.txt” and sort on age using Bubble sort.

Assumed file format (one record per line): Name Age e.g. Raj 30

Answer:

```
#include <stdio.h>

struct Emp
{
    char name[50];
    int age;
};

void bubbleSort(struct Emp e[], int n)
{
    int i, j;
    struct Emp temp;

    for(i = 0; i < n - 1; i++)
    {
        for(j = 0; j < n - i - 1; j++)
        {
            if(e[j].age > e[j + 1].age)
            {
                temp
= e[j];          e[j] = e[j
+ 1];          e[j + 1] =
temp;
            }
        }
    }
}

int main()
{
    FILE *fp, *out;
    struct Emp e[100];    int n
= 0, i;

    fp = fopen("employee.txt", "r");
    if(fp == NULL)
    {
```

```

        printf("File not found");
return 0;
    }

    while(n < 100 && fscanf(fp, "%s %d", e[n].name, &e[n].age) == 2)
n++;    fclose(fp);

    if(n == 0)
{
    printf("No records");
return 0;
}

    bubbleSort(e, n);

    out = fopen("sortedemponage.txt", "w");
if(out == NULL)
{
    printf("Cannot create output file");
return 0;
}

    printf("Sorted data:\n");
for(i = 0; i < n; i++)
{
    fprintf(out, "%s %d\n", e[i].name, e[i].age);
printf("%s %d\n", e[i].name, e[i].age);
}
fclose(out);

    return 0;
}

```

Section II: Database Management Systems [15 Marks]

Q 2) Create the following tables (primary keys are underlined).

Order(orderno, s_order_date, amount)

Client(cno, name, address)

A client can give one or more orders, but an order belongs to exactly one client.

Constraint: Order amount should be > 0

Q 2(A)

A. Create the above database in PostGreSQL and insert sufficient records. [5 marks]

Answer:

```
CREATE TABLE Client(  
  cno INT PRIMARY KEY,  
  name VARCHAR(50),  
  address VARCHAR(100)  
);  
  
CREATE TABLE "Order"(  
  orderno INT PRIMARY  
  KEY,      s_order_date DATE,      amount  
  NUMERIC(12,2) CHECK(amount > 0),      cno INT  
  NOT NULL REFERENCES Client(cno)      ON  
  DELETE CASCADE  
);  
  
INSERT INTO Client VALUES  
(4, 'Amit', 'Pune'),  
(5, 'Neha', 'Mumbai'), (6, 'Raj', 'Nashik');  
  
INSERT INTO "Order" VALUES  
(1, '2019-10-05', 50000, 4),  
(2, '2019-11-10', 20000, 5),  
(3, '2019-10-20', 45000, 4),  
(4, '2019-12-01', 30000, 6),  
(5, '2019-12-15', 41000, 4);
```

Q 2(B)

B. Execute the following queries in PostGreSQL. (Any 3) [6marks]

i. Change order date of client number '04' to '12/4/19'.

Answer:

```
UPDATE "Order"  
SET s_order_date = '2019-04-12' WHERE  
cno = 4;
```

ii. Display all orders given in the month of October.

Answer:

```
SELECT * FROM "Order"  
WHERE EXTRACT(MONTH FROM s_order_date) = 10; iii.
```

Display all orders in descending order of amount.

Answer:

```
SELECT * FROM "Order"  
ORDER BY amount DESC;
```

iv. Delete all the clients of Mumbai.

Q 2(C)

C. Write a query to display the client name who has placed maximum number of orders. [4 marks]

Answer:

```
SELECT C.name  
FROM Client C  
JOIN "Order" O ON C.cno = O.cno  
GROUP BY C.cno, C.name  
HAVING COUNT(*) >= ALL  
(  
    SELECT COUNT(*) FROM "Order"  
    GROUP BY cno  
);
```

Q 3) Viva [5 marks]